**Table 11.1** Percentage of Oxygen

| Oxygen in free state | | Oxygen in combined state | |
|---|---|---|---|
| **Source** | **Percentage** | **Source** | **Percentage** |
| Atmospheric air | 21 % | Plants and animals | 60 – 70 % |
| Water | 88 – 90 % | Minerals in the form of silicates, carbonates and oxides | 45 – 50 % |

$$6CO_2 + 6H_2O \xrightarrow{\text{Energy from the Sun}} C_6H_{12}O_6 + 6O_2$$

In combined state it is present in the earth's crust as silicates and metal oxides. It is also found in water on the surface of the earth. Tri atomic molecule ($O_3$) known as ozone is present in the upper layers of the atmosphere.

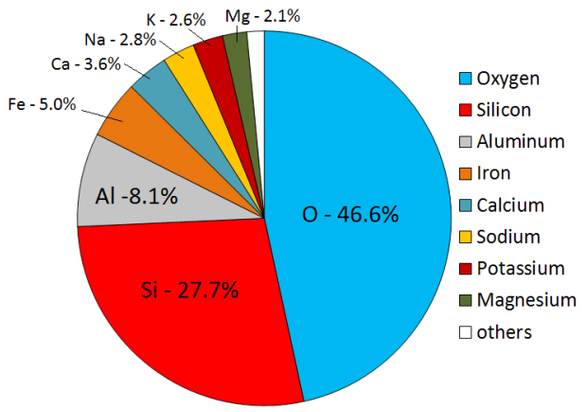



**Figure 11.1** Percentage of elements in the Earth's crust

## 11.1.2 Physical properties of Oxygen

- Oxygen is a colourless, odourless and tasteless gas.
- It is a poor conductor of heat and electricity
- Oxygen dissolves readily in cold water.

DO YOU KNOW?

Oxygen is about two times more soluble in water than nitrogen. If it had the same solubility as nitrogen, then less oxygen would be present in seas, lakes and rivers that will make life much more difficult for living organisms.

- It is denser than air.
- It can be made into liquid (liquified) at high pressure and low temperature.
- It supports combustion.

## 11.1.3 Chemical properties of Oxygen

### 1. Combustibility

Oxygen is a non-combustible gas as it does not burn on its own. But, it supports the combustion of other substances.

DO YOU KNOW?

If oxygen has the capacity to burn itself, striking a match stick will be enough to burn all the oxygen in our planet's atmosphere.

### 2. Reaction with metals

Oxygen reacts with metals like sodium, potassium, magnesium, aluminium, iron etc., to form their corresponding metal oxides which are generally basic in nature. But the metals differ in their reactivity towards oxygen.

$$\text{Metal} + \text{Oxygen} \longrightarrow \text{Metal oxide}$$

**Example**

$$4Na + O_2 \longrightarrow 2Na_2O$$

Sodium Oxygen Sodium oxide

**Activity 1**

Heat a strip of magnesium ribbon in the flame till it catches fire and introduce it into the jar containing oxygen. It burns with a dazzling bright light and white ash of magnesium oxide is formed.



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